AI-Powered Financial Goal Tracking and Advisory Web Application

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DuitWise AI is a web-based application designed to help users manage and track personal financial goals through an interactive dashboard and AI-powered advisory system. The system allows users to monitor savings progress, set financial targets, and receive intelligent guidance through an AI coach interface.

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1. Introduction

1.1 Purpose

The purpose of this report is to describe the analysis, design, implementation and testing of the application named —. The existence of this report allows relevant users to maintain the system, troubleshoot the system, modify and update the system, and view expected functionality.

The intended audience for this report is the team leader and scrum master for the team, the developers and testers for this project, as well as the end users who intend to use the system.

1.2 Scope

The system is intended for individuals who want to manage and track their personal goals through a web application. It is designed for general users who prefer a simple and interactive platform, as well as developers and testers who may maintain or further improve the system in the future. The application allows users to interact with an AI assistant, monitor their progress, and access system features through an easy-to-use interface.

Tech Stack

FRONTEND	Next.js (Tailwind CSS)
DATABASE	MongoDB
AI ENGINE	Google Gemini API
BACKEND	FastAPI python
HOSTING	Vercel or Render

2. Specific Requirements

2.1.1 User Interfaces

The system provides a unified dashboard for users focusing on financial resilience. The main pages include Home (Dashboard), Goal Tracker, AI Money Coach, and Financial Analysis. All core functionalities can be accessed within three clicks from the dashboard.

Required screen formats include interactive progress bars for goal tracking, modal windows for setting financial targets, and a chat-based interface for the AI coach. Error messages appear inline near the affected input field. System notifications appear at the top right of the screen as toast notifications. For example, the system shows a notification when savings are successfully updated.

Standardized page elements such as Save, Cancel, Edit, and Ask buttons appear in fixed locations across all modules. This reduces user confusion and improves navigation consistency.

For action buttons such as Set Goal, Update Savings, and Ask Coach, the buttons are clickable with white text and an Emerald background (#10B981). Data visualization includes the Goal Progress Chart and Spending Pattern Table.

Alerts indicate important financial conditions. For example, the system displays alerts when spending exceeds 80 percent of a budget or when high-risk BNPL debt patterns are detected.

2.1.2 Hardware Interfaces

The DuitWise AI web application can be accessed using desktop and laptop computers. The device must be able to run a modern web browser.

Supported browsers include the latest versions of Google Chrome, Safari, Microsoft Edge, and Mozilla Firefox.

Input devices include a keyboard and mouse or trackpad. These devices are required for interacting with the financial dashboard and AI chat interface.

The system also supports mobile-responsive design using Tailwind CSS. Smartphones and tablets can be used as secondary access devices.

2.1.3 Software Interfaces

Database Management System (DBMS)
Name: MongoDB
Version: 7.0 or later
Purpose: Stores user profiles, financial goals, and transaction summaries.
Interface: Uses the Motor asynchronous MongoDB driver for Python.

Frontend Framework
Name: Next.js (React.js)
Version: 16.1.6
Source: Meta Platforms and Vercel
Purpose: Provides a dynamic user interface with server-side rendering and optimized performance.

Backend Framework
Name: FastAPI
Version: 0.115 or later
Source: Tiangolo
Purpose: Provides REST API endpoints and manages backend application logic.

AI Interface
Name: Ollama or Google Gemini API
Purpose: Generates personalized financial advice and analyzes spending behavior.

2.1.4 Communication Interfaces

HTTP / HTTPS Protocols

The system uses HTTPS for secure communication between the Next.js frontend and FastAPI backend. This protocol is used for operations including authentication, goal updates, and AI advice retrieval.

HTTP Methods

GET retrieves dashboard information, financial goals, and AI advice history.

POST submits new saving goals and AI advice requests.

PUT or PATCH updates existing savings progress and goal details.

DELETE removes financial goals or stored records.

Headers

Authorization uses Bearer tokens to secure API access.

Content-Type uses application/json for standard requests and multipart/form-data for file uploads.

CORS

CORS is configured to allow the Next.js frontend to communicate with the FastAPI backend when hosted on different domains.

For example, the frontend may run on localhost:3000 while the backend runs on localhost:8000 during development.

3. System Architectural Design and Components

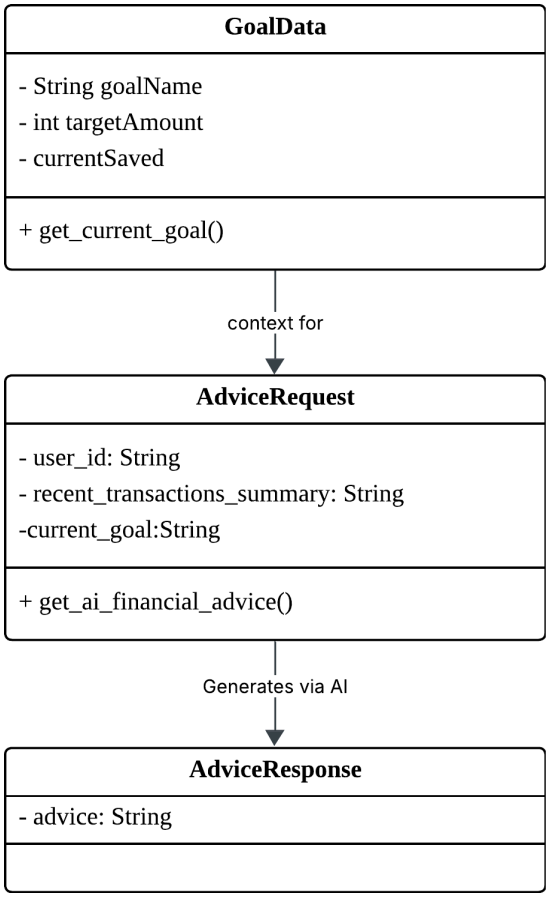

Diagram 3.1: class diagram

The class diagram contains three classes which are GoalData, AdviceRequest, and AdviceResponse. GoalData stores information about the user's financial goal including the goal name, target amount, and the current amount saved. It also has a method to retrieve the current goal. AdviceRequest holds the information needed to request financial advice such as the user ID, a summary of recent transactions, and the current goal, and includes a method to generate AI financial advice. AdviceResponse represents the result returned by the system and stores the generated financial advice. The diagram shows the flow where goal data is used to create an advice request which then produces an advice response.

Diagram 3.2: Package Diagram

The package diagram shows the main structure of the system divided into five packages which are Web App, Creative UI, Economy Module, Intelligence Module, and Core API. The Web App package contains the main layout and Next.js pages that structure the web application. The Creative UI package includes components related to the user interface such as the AI coach chat, savings progress display, and global styles. The Economy Module manages financial goal features including user goal routes, savings models, and goal tracking. The Intelligence Module handles the AI functionality including the AI coach route, Ollama integration, and advice models. The Core API package manages the backend services including the FastAPI application, CORS middleware, and the main router that connects the system components.